

FROZEN FOUNTAIN

ICE STUPAS

Parts of the cold, dry, mountainous Himalayan region of Asia receive only 2–2.7 in (50–70 mm) of rainfall in a year and experience temperatures as low as -22°F (-30°C). While snow remains on the peaks all year round, the lower regions are arid in the summer. Scientists have developed a new way of freezing meltwater into mounds called stupas. In this frozen state, water can be stored at high altitudes for irrigating crops in the nearby fields.

Water store

This towering stupa was built by a group of young engineers in the Leh district of Ladakh, India. It serves as a water reserve. The cone shape means the mass of ice has a relatively small area exposed to direct sunlight. Because of this, it melts more slowly than it would if it was in a flat layer. Because it is hollow, a café has been opened inside.

SUMMER IRRIGATION

Farms in the Himalayan foothills rely on the water that flows down from glaciers to survive, but climate change is making the water supply increasingly erratic. As Earth gets warmer, glaciers are disappearing permanently, and farmers have had to develop new ways of storing water to irrigate crops in the summer.

